

Electric energy cannot be stored but is generated from natural sources and delivered as per the demand. The objective of the control strategy is to generate and deliver power in an interconnected system as economically and reliably as possible while maintaining the voltage and frequency within permissible limits. Changes in real power mainly affect the system frequency while reactive power is mainly dependent on change in voltage. As the system load changes continuously, generation is adjusted automatically to restore the frequency to normal value through automatic generation control (AGC). In an interconnected system consisting of several pools, the role of AGC is to devise the loads among systems, stations and generators and correctly control the scheduled interchanges of the line power while maintaining a uniform frequency.

Now for an interconnected system the fundamental problem is to minimize the source expenses. The production level of each plant should be such that the total cost of generation and transmission is minimum for prescribed schedule of loads. This is called optimal (or economic) load dispatch.

3.1 GENERATOR OPERATING COST

The majority of generators in extants system are of three types—nuclear, hydro and fossile (coal, oil or gases). Nuclear plants tend to be operated at constant output and hydro plants have mainly the capital costs. The total cost of operation includes the fuel cost, costs of labour, supplies and maintenance.

(The operating cost and the limit of power generation of each generator is different. The major component of generator operating cost is the fuel input/hour while maintenance is considered a fixed part of it. Fuel cost is meaningful in case of thermal and nuclear stations but for hydro stations the operating cost is not meaningful. Here we shall concentrate on fuel fired stations. The operating cost of a generator can be expressed in terms of rupees per hour versus output in megawatts.)

(The power output of a thermal plant is increased sequentially by opening a set of valves at the inlet to its steam turbine.) The operating cost of the plant may have the form shown in Fig. 3.1.

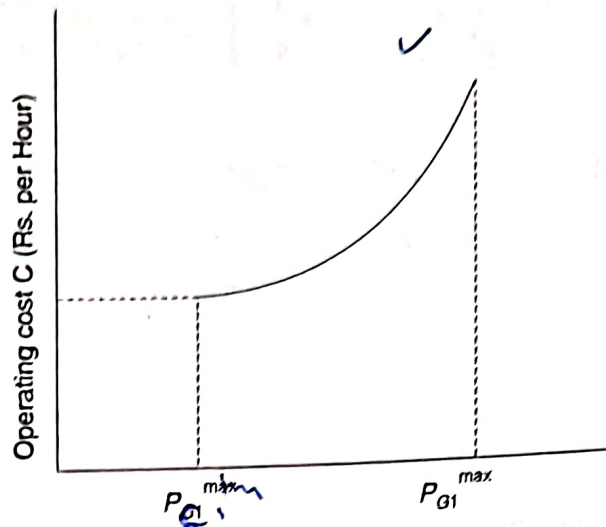


Figure 3.1 Cost curve of the plant

$(P_G^{\min})$ is the upper limit of power generated obtained when throttling is fully opened. $P_G^{\min}$ is the lower limit of power generated when throttling is just opened. The fuel cost curve can be represented by an algebraic equation of n^{th} order by curve fitting technique. In all practical purposes, the fuel cost of generators can be modelled as quadratic equation of real power generation.

$$C = a P_G^2 + b P_G + c$$

3.2 ECONOMIC DISPATCH PROBLEM

Let there be N generators connected either on a bus bar or connected through a transmission network with negligible transmission losses committed to supply an active load equal to P_D .

Then

$$P_{G1} + P_{G2} + P_{G3} + \dots + P_{GN} = P_D \quad (3.1)$$

This is called equality constraint.

Each generator has the lower and upper limit of power generation.

$$P_{Gi}^{\min} < P_{Gi} < P_{Gi}^{\max}$$

This is called inequality constraint.

And the cost function should be optimum for definite set of generating power $(P_{G1}, P_{G2}, \dots, P_{GN})$

$$C = C_1 + C_2 + \dots + C_n \quad (3.2)$$

The operating costs of i^{th} generator are given by n different equation

$$C_i = a_i P_{Gi}^2 + b_i P_{Gi} + c_i \quad (3.3)$$

The cost function C is minimized using Lagrange multiplier method. The Lagrange function is defined as

$$L = C + \lambda(P_D - \Sigma P_{Gi}) \quad (3.4)$$

where λ is the Lagrangian multiplier.

A necessary condition for a function C , to be minimum, is that the partial derivative of the Lagrange function L with respect to each of its arguments must be zero, i.e.,

$$\begin{aligned} \frac{\partial L}{\partial P_{Gi}} = 0 &\Rightarrow 0 = \frac{\partial C_i}{\partial P_{Gi}} + \lambda(0 - 1) \\ \Rightarrow \frac{dC_i}{dP_{Gi}} &= \lambda \\ \frac{\partial L}{\partial \lambda} = 0 &\Rightarrow 0 = P_D - \Sigma P_{Gi} \\ \Sigma P_{Gi} &= P_D \end{aligned} \quad (3.5)$$

The first (3.1) equation represents n different equations and second equation (3.5) is single equation making total $(n + 1)$ equation.

Hence, n unknowns ($P_{G1}, P_{G2}, \dots, P_{Gn}, \lambda$) can be obtained by the solution of these algebraic equations.

$$\begin{aligned} \frac{dC_i}{dP_{Gi}} = \lambda &\Rightarrow 2a_i P_{Gi} + b_i = \lambda \\ P_{Gi} &= \frac{\lambda}{2a_i} - \frac{b_i}{2a_i} \end{aligned} \quad (3.6)$$

Putting the value in $\Sigma P_{Gi} = P_D$

$$\begin{aligned} \Sigma \frac{\lambda}{2a_i} - \Sigma \frac{b_i}{2a_i} &= P_D \\ \lambda \Sigma \frac{1}{2a_i} &= P_D + \Sigma \frac{b_i}{2a_i} \\ \lambda &= \frac{P_D + \Sigma \frac{b_i}{2a_i}}{\Sigma \frac{1}{2a_i}} \end{aligned} \quad (3.7)$$

By putting this value of λ in eq. 3.6 we can obtain $P_{G1}, P_{G2}, \dots, P_{Gn}$. If any of generator is going beyond its minimum or maximum limit its generating power is fixed at that minimum or maximum set point and λ is again obtained among participating generators and total demand (P_D) is assumed equal to earlier demand minus the minimum or maximum set value of non-participating generator (s).

3.3 ECONOMIC DISPATCH INCLUDING TRANSMISSION LOSSES

When transmission distances are very small and load density is very high, transmission losses may be neglected and optimal dispatch of generation is achieved with all plants operating at

equal incremental production cost. The transmission losses may be expressed as a quadratic function of generator powers. The Kron's formula for transmission losses is given by equation.

$$P_L = B_{00} + (B_{01} P_{G1} + B_{02} P_{G2} + \dots B_{0n} P_{Gn}) + \sum_{l=1}^N \sum_{j=1}^N P_{Gi} B_{ij} P_{Gj} \quad (3.8)$$

In the matrix form, it can be written as

$$P_L = B_{00} + [P_{G1} \ P_{G2} \ \dots \ P_{Gn}] \begin{bmatrix} B_{01} \\ B_{02} \\ \vdots \\ B_{0n} \end{bmatrix} + [P_{G1} \ P_{G2} \ \dots \ P_{Gn}] \begin{bmatrix} B_{11} & B_{12} & \dots & B_{1n} \\ B_{21} & B_{22} & \dots & \dots \\ B_{31} & B_{32} & \dots & \dots \\ \vdots & \vdots & \dots & \vdots \\ B_{n1} & \dots & \dots & B_{nn} \end{bmatrix} \begin{bmatrix} P_{G1} \\ P_{G2} \\ \vdots \\ P_{Gn} \end{bmatrix} \quad (3.9)$$

The first two terms in the above equations can be judiciously neglected in most of cases.

$$P_L = \sum_{l=1}^N \sum_j P_{Gi} B_{ij} P_{Gj} = [P_G^T] [B] [P_G] \quad (3.10)$$

Cost function $C = C_1 + C_2 + C_3 + \dots C_n$

In equality constraints $P_{Gi\min} < P_{Gi} < P_{Gi\max}$.

Equality constraints $P_D + P_L = \sum_1^N P_{Gi} \quad (3.11)$

Lagrange function can be defined

$$L = C + \lambda \left(P_D + P_L - \sum_1^N P_{Gi} \right) \quad (3.12)$$

Necessary conditions for optimization is

$$\frac{\partial L}{\partial P_{Gi}} = 0 \text{ and } \frac{\partial L}{\partial \lambda} = 0$$

This leads to

$$\begin{aligned} \frac{\partial C_i}{\partial P_{Gi}} + \lambda \left(\frac{\partial P_L}{\partial P_{Gi}} - 1 \right) &= 0 \\ P_D + P_L - \sum_1^N P_{Gi} &= 0 \\ \frac{\partial C_i}{\partial P_{Gi}} \frac{1}{\left(1 - \frac{\partial P_L}{\partial P_{Gi}} \right)} &= \lambda = L_i \frac{\partial C_i}{\partial P_{Gi}} \end{aligned} \quad (3.13)$$

where $L_i = \frac{1}{\left(1 - \frac{\partial P_L}{\partial P_{Gi}} \right)}$ is called penalty factor.

To obtain the solution, the values of $\frac{\partial C_i}{\partial P_{Gi}}$ and $\frac{\partial P_L}{\partial P_{Gi}}$ are substituted in eq. 3.13.

$$\begin{aligned}
2a_i P_{Gi} + b_i &= \lambda \left(1 - B_{oi} - \sum_{\substack{j=1 \\ j \neq i}}^N 2B_{ij} P_{Gj} \right) \\
&= \lambda \left(1 - B_{oi} - \sum_{\substack{j=1 \\ j \neq i}}^N 2B_{ij} P_{Gj} \right) - \lambda 2B_{ii} P_{Gi} \\
(2a_i + 2\lambda B_{ii}) P_{Gi} &= \lambda \left(1 - B_{oi} - \sum_{\substack{j=1 \\ j \neq i}}^N 2B_{ij} P_{Gj} \right) - b_i \\
P_{Gi} &= \frac{\lambda \left(1 - B_{oi} - \sum_{\substack{j=1 \\ j \neq i}}^N 2B_{ij} P_{Gj} \right) - b_i}{2(a_i + \lambda B_{ii})} \quad (3.14)
\end{aligned}$$

If the initial values of $P_{Gi} (i = 1, 2, \dots, N)$ and λ are known, the above equation can be solved iteratively until equation of equality constraints is satisfied subjected each generator remains within specific limits.

For certain estimated value of λ^D , P_{G1} , P_{G2} , ..., P_{Gn} and P_L are obtained. This should satisfy the equality constraint.

$$\Sigma P_{Gi}^o - P_D + P_L^o = \Delta P \quad (3.15)$$

$$\Sigma \frac{\lambda \left(1 - B_{io} - \sum_{\substack{j=1 \\ j \neq i}}^N 2B_{ij} P_{Gj} \right) - b_i}{2(a_i + \lambda B_{ii})} = P_D + P_L \quad (3.16)$$

$$f(\lambda) = P_D + P_L \quad \text{let } \lambda = \lambda^0 + \Delta\lambda^0$$

$$f(\lambda^{(0)} + \Delta\lambda^{(0)}) = P_D + P_L^{(0)} \quad \lambda = \text{actual value of } \lambda$$

$$\lambda^{(0)} = \text{initial estimation}$$

$$\Delta\lambda^0 = \text{error}$$

Apply Taylor's series and neglecting higher order terms.

$$f(\lambda)^{(0)} + \left(\frac{df}{d\lambda} \right)^0 \Delta\lambda^0 = P_D + P_L^{(0)}$$

$$\Sigma P_{Gi}^0 + \left(\frac{df}{d\lambda} \right)^0 \Delta\lambda^0 = P_D + P_L^{(0)} \quad (3.17)$$

$$\Delta\lambda^0 = \frac{\Delta P}{\left(\frac{df}{d\lambda}\right)^0}$$

Using formula

$$\frac{d}{d\lambda} \left(\frac{A}{B} \right) = \frac{\frac{dA}{d\lambda} - \frac{AdB}{d\lambda}}{B^2}$$

$$\frac{df}{d\lambda} = \frac{\frac{1}{2} \sum \left[(a_i + B_{ii}\lambda) \left(1 - B_{io} - \sum_{j \neq i} 2B_{ij}P_{Gj} \right) - B_{ii} \right] \left[\lambda \left(1 - B_{io} - \sum_{j \neq i} 2B_{ij}P_{Gj} \right) - b_i \right]}{(a_i + \lambda B_{ii})^2}$$

$$\frac{df}{d\lambda} = \sum_{i=1}^N \frac{a_i \left[(1 - B_{io}) - \sum_{j \neq i} 2B_{ij}P_{Gi} \right] + B_{ii}b_i}{2(a_i + \lambda B_{ii})^2}$$

$$\lambda^1 = \lambda^0 + \Delta\lambda^0$$

(3.18)

The iterations are repeated.

The problem solving technique through iteration method is presented in flow chart on page 65.

3.4 TRANSMISSION LOSS FORMULA DERIVATION

Transmission loss P_L is expressed without loss of accuracy as a function of real power generations.

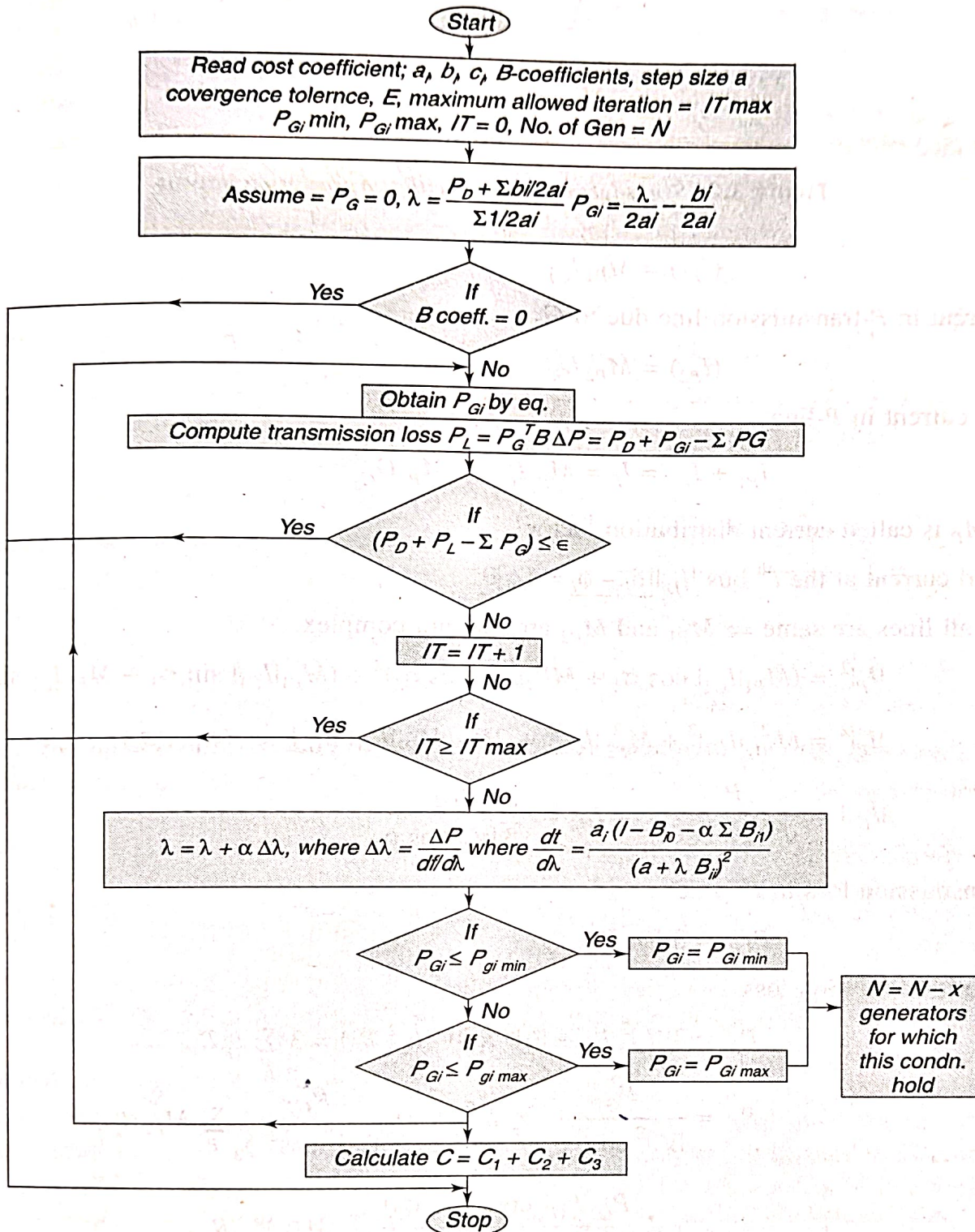
The power loss is expressed using B -coefficients called loss coefficients.

Expression for transmission power loss is derived using KRON's method of reducing system to an equivalent system with a single hypothetical load.

The expression is based on several assumptions as follows:

- (i) All the lines in the system have same $\frac{X}{R}$ ratio.
- (ii) All the load currents have the same phase angle.
- (iii) All the load currents maintain a constant ratio to the total current.
- (iv) The magnitude and phase angle of bus voltages at each station remains constant.
- (v) Power factor at each station bus remains constant.

Let there be two generating stations supplying an arbitrary number of loads through a transmission network as shown in Fig. 3.2.



To determine the current in any line say P^{th} line, apply superposition principle and determine the current passing through the line, I_P .

Current in P transmission line due to G_1 .

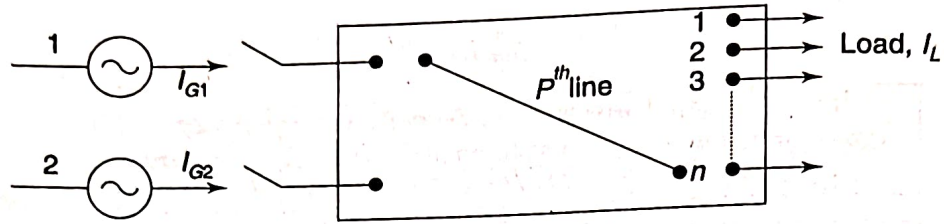


Figure 3.2 Transmission network with two generating stations

$$(I_{P1}) = M_{P1} I_{G1}$$

Current in P -transmission line due to G_2

$$(I_{P2}) = M_{P2} I_{G2}$$

Net current in P -line

$$I_{P1} + I_{P2} = I_P = M_{P1} I_{G1}^{\angle \sigma_1} + M_{P2} I_{G2}^{\angle \sigma_2}$$

where M_P is called current distribution factor

$$\text{Load current at the } i^{\text{th}} \text{ bus } |I_{Di}| |\delta_i - \phi_i| = I_{Di} |\theta|$$

ratio of all lines are same $\Rightarrow M_{P1}$ and M_{P2} are real not complex.

$$|I_P|^2 = (M_{P1} |I_{G1}| \cos \sigma_1 + M_{P2} |I_{G2}| \cos \sigma_2)^2 + (M_{P1} |I_{G1}| \sin \sigma_1 + M_{P2} |I_{G2}| \sin \sigma_2)^2$$

$$|I_P|^2 = M_{P1}^2 |I_{G1}|^2 + M_{P2}^2 |I_{G2}|^2 + 2M_{P1} M_{P2} |I_{G1}| |I_{G2}| \cos (\sigma_1 - \sigma_2)$$

$$|I_{G1}| = \frac{P_{G1}}{\sqrt{3} |V_1| \cos \phi_1} = I_{G2} = \frac{P_{G2}}{\sqrt{3} |V_2| \cos \phi_2}$$

Transmission loss in P^{th} line

$$P_{LP} = 3 |I_P|^2 R_P$$

Total transmission loss

$$P_L = 3 [|I_1|^2 R_1 + |I_2|^2 R_2 + |I_3|^2 R_3] = 3 \sum_P I_P^2 R_P$$

$$P_L = \frac{P_{G1}^2}{|V_1|^2 \cos^2 \phi_1} \sum_P M_{P1}^2 R_P + \frac{P_{G2}^2}{|V_2|^2 \cos^2 \phi_2} \sum_P M_{P2}^2 R_P$$

$$+ 2 \frac{P_{G1} P_{G2} \cos(\sigma_1 - \sigma_2)}{|V_1| |V_2| \cos \phi_1 \cos \phi_2} \sum_P M_{P1} M_{P2} R_P \quad (3.19)$$

$$P_L = P_{G1}^2 B_{11} + 2 P_{G1} P_{G2} B_{12} + P_{G2}^2 B_{22}$$

$$= [P_{G1} \ P_{G2}] \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} \begin{bmatrix} P_{G1} \\ P_{G2} \end{bmatrix} = P_G^T B P_G \quad (3.20)$$

Equation 3.20 expresses the total loss as a function of real power generations P_{G1} and P_{G2} . The coefficients B_{11} , B_{22} and B_{12} are called loss coefficients (or) B -coefficients and the unit is $(\text{MW})^{-1}$ and also considered to be constant in view of assumptions made. The same procedure can be extended for systems having more number of stations. If the system has ' n ' number of stations, supplying the total load through transmission lines, the transmission line loss is given by,

$$P_L = \sum_{i=1}^n \sum_{j=1}^n P_{Gi} B_{ij} P_{Gj} \quad (3.21)$$

When $n = 2$,

$$\begin{aligned} P_L &= \sum_{i=1}^2 \sum_{j=1}^2 P_{Gi} B_{ij} P_{Gj} \\ &= P_{G1} B_{11} P_{G1} + P_{G1} B_{12} P_{G2} + P_{G2} B_{21} P_{G1} + P_{G2} B_{22} P_{G2} \\ &= B_{11} P_{G1}^2 + B_{22} P_{G2}^2 + 2B_{12} P_{G1} P_{G2} \end{aligned}$$

Similarly, for $n = 3$,

$$\begin{aligned} P_L &= \sum_{i=1}^3 \sum_{j=1}^3 P_{Gi} B_{ij} P_{Gj} \\ &= B_{11} P_{G1}^2 + B_{22} P_{G2}^2 + B_{33} P_{G3}^2 + 2B_{12} P_{G1} P_{G2} + 2B_{23} P_{G2} P_{G3} + 2B_{31} P_{G3} P_{G1} \end{aligned} \quad (3.22)$$

Since the transmission lines are symmetrical, loss coefficients B_{ij} and B_{ji} are equal, i.e., B_{ij} coefficients are the loss coefficients and represented in matrix form of n generator system as

$$B = \begin{bmatrix} B_{11} & B_{12} & \dots & B_{1n} \\ B_{21} & B_{22} & \dots & B_{2n} \\ \dots & \dots & \dots & \dots \\ B_{n1} & B_{n2} & \dots & B_{nn} \end{bmatrix}_{n \times n}$$

The diagonal elements of these coefficients are all positive and strong (since generating stations are interconnected) as compared with the off diagonal elements which mostly are negative and relatively weaker.

These coefficients are determined for a large system by an elaborate digital computer program starting from the assembly of the open circuit impedance matrix of the transmission line which is quite lengthy and time consuming. Besides, the formulations of B -coefficients are based on several assumptions and do not take into account the actual conditions of the system, the solution for the plant generations cannot be expected to be the best for minimum cost of generation.

3.5 OPTIMAL HYDROTHERMAL SCHEDULING

In practice, the power system consists of hydro and thermal power stations both. There is need of economical loading of integrated system. The operating cost of a thermal plant is very high and its capital cost is low. On the other hand, the operating cost of hydro electric plant is low and its capital cost is high. In a hydro thermal integration, it is essential to use the total quantity of water available from the hydro system to the fullest extent. In a hydro system, fixed charges continue regardless of power generation, as there is no fuel cost. Therefore minimum overall cost is achieved by maximum exploitation of hydro resources over a period of time. Several hydro electric plants are multipurpose. Therefore, it is needed to meet certain obligations other than power generation. These may include minimum plant discharge to meet irrigation and navigational commitments, limit to maximum discharge against danger of flood.

The output of hydro unit varies with effective heads and rate of water discharge for large capacity reservoir, it is practical to assume constant effective head. Problem can be formulated for the following cases:

- (1) Short range fixed head hydro thermal scheduling.
- (2) Short range variable head hydro thermal scheduling.
- (3) Long range generation scheduling of hydro thermal system.

3.6 CLASSIFICATION OF HYDRO PLANTS

Classification on the Basis of Type

(Hydro power plants, on the basis of their type, are classified into pumped storage plants and conventional plants. The conventional plants are further classified into storage plants and run-of-river plants.

Pumped storage plants. A pumped storage plant is associated with upper and lower reservoirs. During light load periods, water is pumped from the lower to the upper reservoirs using available energy from other sources as surplus energy. During peak load the water stored in the upper reservoir is released to generate power to save fuel costs of thermal plants.) The pump storage plant is operated until the added pumping cost exceeds the savings in thermal costs due to the peak sharing operation.

(**Conventional plants.** Conventional plants are classified into run-of-river plants and storage plants.

Run-of-river plants have little storage capacity, and use water as it becomes available. Water not utilized is spilled.

Storage plants are associated with reservoirs which have significant storage capacity. During periods of low power requirements, water can be stored and then released when the demand is high.)

Classification on the Basis of Location

On the basis of their location, hydro plants are classified into three different categories:

- (a) Hydro plants on different streams
- (b) Hydro plants on the same stream (serial or cascaded plants)
- (c) Multi-chain hydro plants.

Hydro plants on different streams. These hydro plants are located on different streams and are independent of each other as shown in Fig. 3.3

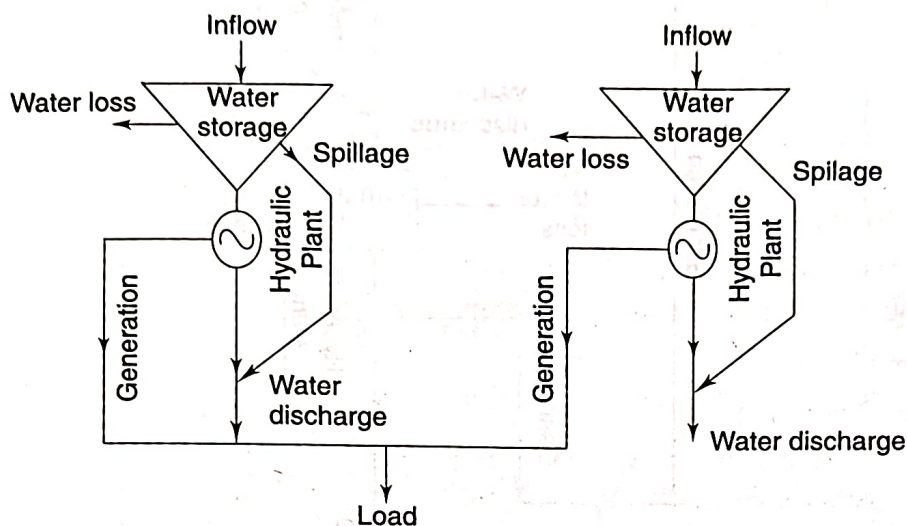


Figure 3.3 *Hydro plants on different streams*

Hydro plants on the same stream (serial or cascaded plants). When hydro plants are located on the same stream, the downstream plant depends on the immediate upstream plant. The arrangement is shown in Fig. 3.4. The downstream plant influences the immediate upstream plant by its effect on tail water elevation and effective head.

Multi-chain hydro plants. These hydro plants are located on different streams as well as on the same stream. The arrangement is shown in Fig. 3.5.

'Further, in view of the increase in the demand for electricity for purposes such as industrial, agricultural, commercial and domestic together with the high cost of fuel as well as its limited reserve, considerable attention is being given to hydrothermal coordination problem. Besides the insignificant incremental cost involved in hydro generation, the problem of minimizing the operational cost of a hydrothermal system can be reduced essentially to that of minimizing the fuel cost for thermal plants under the constraints of the water available for hydro generation in a given period of time. The cyclic nature of water flows and load demand, as well as the validity of model assumptions, suggest splitting the problem into long and short periods.

3.7 LONG-RANGE PROBLEM

The long-range problem considers the yearly cyclic nature of reservoir water inflows and seasonal load demand and correspondingly a scheduling period of one year is used. The solution of the long-range problem considers the dynamics of head variations through the water flow continuity equation. The long-term hydrothermal scheduling problem can be classified in three groups:

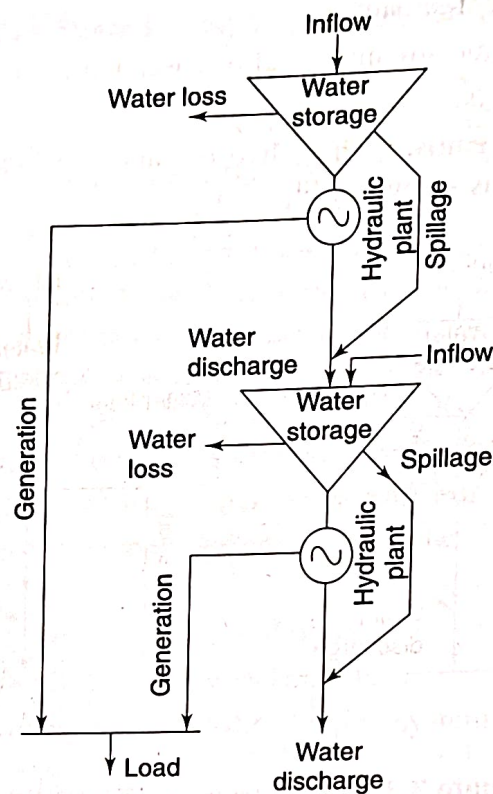


Figure 3.4 Hydro plants on the same streams

Multi-storage hydroelectric systems

These hydro plants are located on different streams and have their independent reservoirs.

Cascaded hydroelectric systems

These hydro plants are located on the same streams and the downstream plant depends on the immediate upstream plant. The upstream plant inputs discharge to the immediate downstream plant.

Multi-chain hydroelectric systems

These hydro plants are located on different streams as well as follow the same stream.

3.8 SHORT-RANGE PROBLEM

Here, the load demand on the power system exhibits cyclic variation over a day or a week and correspondingly the scheduling interval for short range is either a day or a week. As the scheduling interval of short-range problem is small, the solution of the short-range problem can assume the head to be fairly constant. The amount of water to be utilized for the short-range scheduling problem is known from the solution of the long-range scheduling problem. The short-term hydrothermal scheduling problem is classified into two groups:

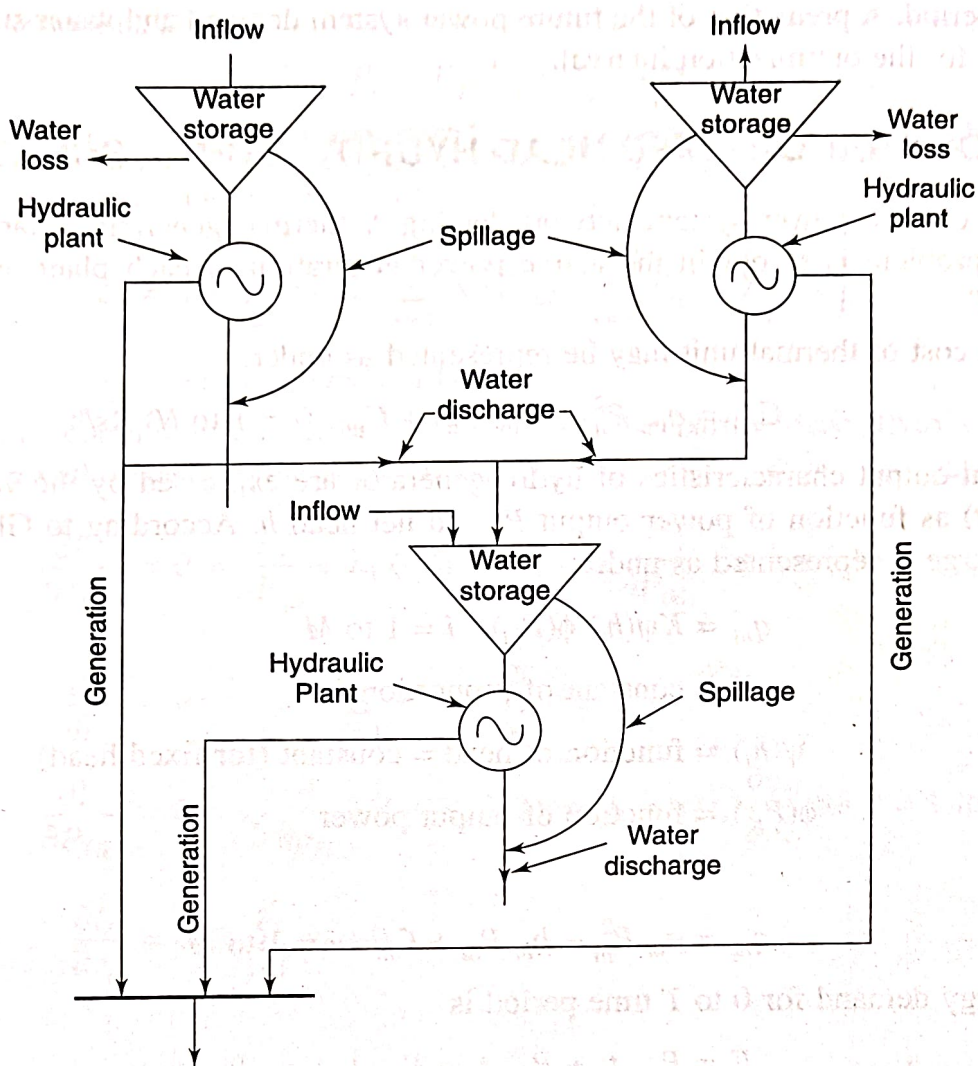


Figure 3.5 Multi-chain hydro plants

Fixed-head hydrothermal scheduling

The head of reservoir can be assumed fixed if the hydro plants have reservoirs of large capacity.

Variable-head hydrothermal scheduling

The head of reservoir is variable if the hydro plants are having reservoirs of small capacity.

The short-range or long-range hydrothermal scheduling problem can also contain pumped storage plants. During light load period, water is pumped from the lower to upper reservoirs using the available energy from other sources as surplus energy. During peak load, water stored in the upper reservoir is released to generate power to save fuel costs of thermal plants.

The main objective of hydrothermal operation is to minimize the total system operating cost represented by the fuel cost required for the system's thermal generation, over the optimization interval. Each hydro plant is constrained by the amount of water available for draw-down in the

scheduling period. A prediction of the future power system demand and water supply is assumed to be known for the optimization interval.

3.9 SHORT-RANGE FIXED HEAD HYDROTHERMAL SCHEDULING

Consider an electric power system network having N thermal generating plants and M hydro plants. The problem is to obtain the active power generation of each plant over a finite time period 0 to T .

The fuel cost of thermal unit may be represented as under:

$$C_{thi} = a_{thi} P_{thi}^2 + b_{thi} P_{thi} + C_{thi} \quad (i = 1 \text{ to } N) \text{ Rs/h} \quad (3.23)$$

The input-output characteristics of hydro generator are expressed by the variation of water discharge $q(t)$ as function of power output P_h and net head h . According to Glimn-Kirchmayer model discharge is represented as under:

$$q_{hi} = K\psi(h_i) \phi(P_{hi}) \quad i = 1 \text{ to } M$$

K = constant of proportionality

$\psi(h_i)$ = function of head = constant (for fixed head)

$\phi(P_{hi})$ = function of output power

Thus

$$q_{hi} = a_{hi} P_{hi}^2 + b_{hi} P_{hi} + C_{hi} \quad i = 1 \text{ to } M \quad (3.24)$$

Net energy demand for 0 to T time period is

$$E = P_{D1} t_1 + P_{D2} t_2 + P_{D3} t_3 + \dots P_{Dn} t_n \quad (3.25)$$

$$T = t_1 + t_2 + t_3 + \dots t_n \quad (3.26)$$

Now objective is to minimize the following function

$$\begin{aligned} C = & (t_1 C_{11} + t_1 C_{21} + t_1 C_{31} + \dots) + (t_2 C_{12} + t_2 C_{22} + t_2 C_{32} + \dots) \\ & + \dots (t_k C_{1k} + t_k C_{2k} + t_k C_{3k} + \dots) \dots \\ & + (t_n C_{1n} + t_n C_{2n} + t_n C_{3n} + \dots t_n C_{4n}) \end{aligned} \quad (3.27)$$

where C_{ik} = cost of i^{th} generator at k^{th} interval of load, generating P_{thik} power subjected the following equality constraints

$$P_{th1k} + P_{th2k} + P_{th3k} + \dots P_{thnk} + P_{h1k} + P_{h2k} + \dots P_{hmk} = P_{Dk} \text{ for } k = 1 \text{ to } n \quad (3.28)$$

$$t_1 q_{h1} + t_2 q_{h2} + t_3 q_{h3} + t_k q_{hik} + t_n q_{h_n} = V_i =$$

Net water discharged by i^{th} hydro plant during complete period (T)

$$i = 1 \text{ to } M$$

(3.29)

Frequently constraints are

$$P_{hi}^{\min} < P_{hi} < P_{hi}^{\max} \quad i = 1 \text{ to } M$$

$$P_{thi}^{\min} < P_{thi} < P_{thi}^{\max} \quad i = 1 \text{ to } N$$

The above objective is augmented by the constraints:

$$L = C + \sum_{i=1}^M v_i V_i + \sum_{k=1}^n \lambda_k \left(P_{DK} - \sum_{i=1}^N P_{thi} - \sum_{i=1}^M P_{hi} \right) \quad (3.30)$$

v_i = water conversion factor

The optimality conditions are described by taking the partial derivatives of augmented objective function with respect to the decision variables P_{thi} , P_{hi} , λ_k & v_i

$$\frac{\partial L}{\partial P_{thik}} = 0 = \frac{\partial C}{\partial P_{thi}} + \lambda_k (-1) \Rightarrow \lambda_k = t_k \frac{\partial C_i}{\partial P_{thik}} \text{ for } k = 1 \text{ to } n$$

$$\frac{\partial L}{\partial \lambda_k} = 0 \Rightarrow P_{DK} = \sum_{i=1}^N P_{thik} + \sum_{i=1}^M P_{hik}$$

$$\frac{\partial L}{\partial P_{hik}} = 0 = v_i \frac{\partial}{\partial P_{hi}} \Sigma V_i + \lambda_k (-1) \Rightarrow \lambda_k = v_i t_k \frac{\partial q_{hi}}{\partial P_{hi}} \text{ for } k = 1 \text{ to } n$$

$$\frac{\partial L}{\partial v_i} = 0 \Rightarrow \sum_{i=1}^M v_i = 0$$

Rewriting

$$P_{DK} = (P_{th1k} + P_{th2k} + \dots P_{thNk}) + (P_{h1k} + P_{h2k} + \dots P_{hMk}) \text{ for } k = 1 \text{ to } n \quad (3.31)$$

$$t_1 q_{h_{i1}} + t_2 q_{h_{i2}} + t_3 q_{h_{i3}} + \dots t_k q_{h_{ik}} + \dots t_n q_{h_{in}} = V_i, \quad i = 1 \text{ to } M \quad (3.32)$$

$$\lambda_k = t_k (2a_{thi} P_{thik} + b_{thi}), \quad K = 1 \text{ to } n \quad (3.33)$$

$$\lambda_k = v_i t_k (2a_{hi} P_{hik} + b_{hi}), \quad K = 1 \text{ to } n \quad (3.34)$$

Unknown variables are

$$\begin{array}{l} P_{thik} \left\{ \begin{array}{l} k = 1 \text{ to } n \\ i = 1 \text{ to } N \end{array} \right\} \\ P_{hik} \left\{ \begin{array}{l} k = 1 \text{ to } n \\ i = 1 \text{ to } m \end{array} \right\} \\ \lambda_k \left\{ \begin{array}{l} k = 1 \text{ to } n \end{array} \right\} \end{array} \quad i = 1 \text{ to } N$$

$$q_{hik} \left\{ \begin{array}{l} k = 1 \text{ to } n \\ i = 1 \text{ to } M \end{array} \right\}$$

$$v_i \quad i = 1 \text{ to } M$$

Total unknown variables

$$nN + nM + n + nM + M = nN + 2nM + n + m$$

Known variables

$$P_{DK} \quad k = 1 \text{ to } n$$

$$t_i \quad i = 1 \text{ to } n$$

$$v_i \quad i = 1 \text{ to } M$$

a_{thi} , b_{thi} , a_{hi} , and b_{hi} are constants.

Example 3.1 The cost curves of two generators are as follows

$$C_1 = 0.1 P_{G1}^2 + 40P_{G1} + 120 \text{ Rs/hr}$$

$$C_2 = 0.125 P_{G2}^2 + 30P_{G2} + 100 \text{ Rs/hr}$$

And the daily load cycle of the system is given in Fig. 3.6.

Consider the 24 hours period from 6 AM one morning to 6 AM next morning. Find out whether it would be more economical to keep both the units in service for this 24 hours period or to remove one of the units from service for the 12 hours of light load. Assume that a cost of Rs. 400 is incurred in taking either unit off the line and returning it to service after 12 hours. The maximum and minimum loads on unit is to be 125 MW and 20 MW respectively.

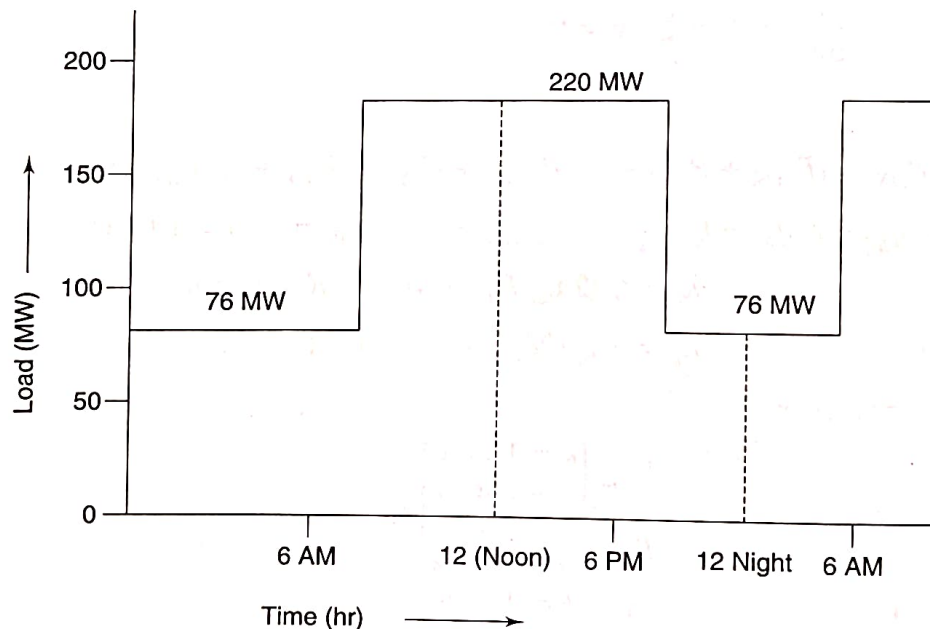


Figure 3.6

Solution: Case 1: When both units are operating throughout 24 hr

Total time = 24 hr (consider)

Total load = 220 MW for 12 hr + 76 MW for 12 hr

(6 AM to 6 PM)

(6 AM to 6 AM)

Heavy load: For the 12 hr period, load = 220 MW

i.e., $P_D = 220 \text{ MW}$

We have to find out the optimal scheduling of two units with this load.

We have the cost curves of two units

For unit 1, $C_1 = 0.1 P_{G1}^2 + 40P_{G1} + 120 \text{ Rs/hr}$

Incremental fuel cost, $\frac{dC_1}{dP_{G1}} = 0.1 \times 2P_{G1} + 40$
 $= 0.2 P_{G1} + 40$

For unit 2

$$C_2 = 0.125 P_{G2}^2 + 30P_{G2} + 100 \text{ Rs/hr}$$

$$\Rightarrow \frac{dC_2}{dP_{G2}} = 0.125 \times 2P_{G2} + 30$$

$$= 0.25 P_{G2} + 30$$

For the optimal distributions of load,

$$\frac{dC_1}{dP_{G1}} = \frac{dC_2}{dP_{G2}} = \lambda$$

$$0.2 P_{G1} + 40 = 0.25 P_{G2} + 30$$

$$0.2 P_{G1} - 0.25 P_{G2} = -10 \quad (1)$$

$$P_{G1} + P_{G2} = 220 \quad (2)$$

By solving equations (1) and (2)

$$P_{G1} = 100 \text{ MW}$$

and

$$P_{G2} = 120 \text{ MW}$$

Total fuel cost for this 12 hr period (i.e., 6 AM to 6 PM) with this optimal distribution is

$$= [(0.1 \times 100^2 + 40 \times 100 + 120) + (0.125 \times 120^2 + 30 \times 120 + 100)] \times 12$$

$$= \text{Rs. } 1,27,440/-$$

For light load of 76 MW: If both units operate in the light load period (76 MW from 6 PM to 6 AM) also, we have to find out the optimal system operation

i.e., $0.2P_{G1} + 40 = 0.25P_{G2} + 30$

$$0.2P_{G1} - 0.25P_{G2} = -10 \quad (3)$$

$$P_{G1} + P_{G2} = 76 \quad (4)$$

By solving equations (3) and (4)

$$P_{G1} = 20 \text{ MW}; P_{G2} = 56 \text{ MW}$$

Total fuel cost for this period is

$$\begin{aligned} &= [(0.1) \times (20)^2 + 40 \times 20 + 120] + (0.125 \times 56^2 \\ &\quad + 30 \times 56 + 100) \times 12 \\ &= \text{Rs. } 37,584 \end{aligned}$$

Hence, the total fuel cost when both the units are operating throughout the 24 hr period =
Rs. 1,27,440 + Rs. 37,584

$$= \text{Rs. } 1,65,024$$

Case 2: If only one of the units is run during the light load period, it is easily verified that it is economical to run unit 2 and to put off unit 1.

Then the fuel cost during this period will be

$$\begin{aligned} &= [0.125 \times 76^2 + 30 \times 76 + 100] \times 12 \\ &= \text{Rs. } 37,224 \end{aligned}$$

$$\therefore \text{Total fuel cost for this case} = \text{Rs. } 1,27,440 + \text{Rs. } 37,224 = \text{Rs. } 1,64,664$$

Considering a cost of Rs. 400 is incurred in taking either unit off the line and returning it to service after 12 hr, then total operating cost for this case will be the total fuel cost plus the start-up cost of unit 1.

i.e., total operating cost = total fuel cost + start-up cost

$$= \text{Rs. } 1,64,664 + 400$$

$$= \text{Rs. } 1,65,064$$

Now, Rs. 1,65,064 > Rs. 1,65,024

(for case 2)

(for case 1)

It is clear that it is economical to run both the units.

Example 3.2 Consider the following three I_C curves

$$P_{G1} = -100 + 50(IC_1) + 2(IC_1)^2$$

$$P_{G2} = -150 + 60(IC_2) - 2.5(IC_2)^2$$

$$P_{G3} = -80 + 40(IC_3) - 1.8(IC_3)^2$$

where IC 's are in Rs/MWh and P_G 's are in MW

The total load at a certain hour of the day is 400 MW. Neglect transmission losses and obtain the optimum generation scheduling within an accuracy of ± 0.05 MW.

Note: All P_G 's must be real and positive.